MILITARY SPECIFICATION

CHASSIS, TRAILER: 1 to 5 TON, 4 WHEEL

GENERAL PURPOSE

This specification is approved for use by all
Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers 4 wheeled chassis, trailers, intended for use as tactical vehicles by the United States Armed Forces.

1.2 Classification. Chassis, trailer shall be of the following model, as specified (see 6.2).

Model	Description
MB-1	Chassis, Trailer, 1-Ton, 4-Wheel
XM113E2	Chassis, Trailer, 3-Ton, 4-Wheel
XM456	Chassis, Trailer, 5-Ton, 4-Wheel

2. APPLICABLE DOCUMENTS

2.1 GOVERNMENT DOCUMENTS

2.1.1 Specification, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial Comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMIRFW Robins AFB, GA 31098 - 5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2330

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SPECIFICATIONS

Federal

VV-B-680	Brake Fluid, Automotive

Military

MIL-P-514	Plates, Identification, Transportation Data and Blank.
MIL-L-3150	Lubricating Oil, Preservative, Medium.
MIL-H-6083	Hydraulic Fluid, Petroleum Base, Preservative.
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements For.
MIL-G-10924	Grease, Automotive and Artillery.
MIL-H-13910	Hydraulic fluid, Non-petroleum Base, Automotive Brake, Arctic.
MIL-G-23827	Grease, Aircraft and Instruments, Gear and Actuator Screw.

STANDARDS

Federal

FEDERAL STD: No. 595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-193	Painting Procedures Tactical Vehicles (Tracked and Wheeled).
MIL-STD-281	Automobiles, Trucks, Truck - Tractors, Trailers and Trailer Dollies, Preservation and Packaging of.
MIL-STD-642	Identification Marking of Administrative Combat and Tactical Transport Vehicles.

2.1.2 Other Government Documents, Drawings, and Publications. The following other Government documents, drawings, and publications form a part of this specification to the extend specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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DRAWING

Air Force

54J6207	Trailer Chassis, 2000-Pound Capacity, 4-Wheel, Type MB-1.

Engineering Parts List (EPL)

EPL8750602	Chassis, Trailer, 3-Ton, 4-Wheel, XM113E2
EPL8736219	Chassis, Trailer, 5-Ton, 4-Wheel, XM456

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition function should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2 and 6.9).

3.2 Materials. The materials shall be specified herein and in the referenced specifications and drawings (see 2.1). Material shall be free from all defects and imperfections that might affect the serviceability, utility and appearance of the finished product.

3.3 Designs and construction. Chassis design and construction shall be in accordance with the drawings or drawing listed in their respective Engineering Parts List (see 2.1), except as modified and supplemented herein and by the procuring activity.

3.4 Performance. Chassis shall meet performance requirements of this specification when fully equipped, loaded with applicable payload and serviced with products specified in table I. Vehicles shall be capable of operation as specified herein, under any extreme condition of weather and altitude, in ambient air temperatures ranging from minus 65°F to plus 125°F and when coupled to the tactical prime mover (see 6.5).

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Table I. Service Products Specifications

PRODUCT USE	Ambient air temperature	
	-65°F to 0°F	+10°F To 132°F
Oil:		
For hydraulic brakes	MIL-H-13910	VV-B-680
For shock absorbers	MIL-H-6083	MIL-H-6083
For general purpose lubrication	MIL-L-3150	MIL-L-3150
Grease:		
For sealed bearings	MIL-G-23827	MIL-G-23827
For general chassis, lubrication, including wheel bearings	MIL-G-10924	MIL-G-10924

3.4.1 Payload, towing speeds, and trailing ability.

3.4.1.1 Highway. When concurrently transporting the applicable payload (see table II) uniformly distributed over load area and being towed over level, smooth, improved and prepared hard surfaced roads at the appropriate sustained speed (see table II), the trailer shall track the towing vehicle without weaving from side to side of the path of the towing vehicle moving in a straight line and without swaying laterally to the extent which adversely affects the controllability of the vehicle combination.

3.4.1.2 Cross-country. When simultaneously carrying the applicable maximum rated payload (see table II) uniformly distributed, and being towed over unimproved roads, trails, open fields, rolling hills, and cross-country terrain at the appropriate speed (see table II) the trailer shall follow the prime mover without damage to itself or to the primer mover.

Table II

Vehicle	Payload (Pounds)		Miles per hour	
	Highway	Cross-country	Highway	Cross-country
MB-1	2000	2000	50	15
XM113E2	9620	9620	50	15
XM456	11620	11620	50	15

3.4.2 Slope operation. Vehicle shall be capable of operating without stalling, slipping or upsetting on ascending or descending longitudinal slopes up to 40 percent grade and side slopes up to 20 percent grade.

3.4.3 Brake holding ability.

3.4.3.1 Parking brake. Parking brakes shall conform to MIL-M-8090.

3.4.4 Turning ability. The vehicle shall trail the towing vehicle in turning within its minimum turning circle, without cramping, side-slipping or damge to the vehicle or towing vehicle.

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3.4.5 Fording ability. The vehicle shall tow through fresh or salt water at a depth great enough to be entirely submerged for periods up to 15 minutes duration without damage to the vehicle or the components. This shall be accomplished without the use of special kits and with a minimum of preparation or servicing before and after the fording operation.

3.4.6 Wheel and tire balancing. Wheels, tubes, tires and brake drums shall be effectively balanced in conformance to applicable drawings or as specified in procurement documents. After mounting, each assembly of tire, tube and wheel shall be balanced within practicable limits, so as to eliminate wheel tramp or drumming at road speed of 50 mph.

3.4.7 Extreme climatic operation. Each type of vehicle shall operate under any condition of inclement weather and at ambient temperatures ranging from minus 65°F up to 125°F encountered in frigid, temperate, and tropical zones; and shall withstand, at the extreme conditions of weather and temperatures specified, protracted storage periods without indication of deterioration.

3.4.8 Reliability. Each type of vehicle shall be capable of operating at least 2560 miles in a military environment without failure due to workmanship or materials; and when subjected to military operating conditions specified herein (see 3.5.1 through 3.5.7), shall exhibit flexural stiffness and torsional rigidity to withstand bending and twisting without evidence of failures.

3.5 Treatment and painting. Painting and preparation therefore, shall be in accordance with MIL-STD-193. Colors shall conform to appropriate color chips in Fed. Std. No. 595, and shall be as specified by procuring activity (see 6.2).

3.6 Marking. Registration and other markings shall be in accordance with MIL-STD-642 or as specified by the procuring activity.

3.7 Name, shipping and service data plates. Data plates shall be photographic aluminum plate (sensitized). The plates shall have the photosensitive silver compounds imbedded within the oxide layer; and after photographic processing, the letters, numbers and all other characters shall be sealed in the oxide layer by suitable chemical treatment of the layer. The plates shall have a black background, with white letters and letter blanks, except that caution and warning plates shall have red letters and letter blanks. The size of the plate, type and size of lettering, and the information to be included shall be in accordance with applicable drawings or as specified by the procuring activity. Thickness of the plates shall be 0.030-inch minimum. The plates shall meet the performance requirements of Specification MIL-P-514.

3.8 Servicing and adjusting. Prior to acceptance of the vehicles by the Government inspector, contractor shall adjust and service each vehicle for immediate operational use including at least the following: adjust brake system, check electrical system; inflate all tires; and completely lubricate with grades of lubricants recommended for the ambient air temperature at the delivery point.

3.9 Workmanship. Workmanship shall be of a quality to assure delivery of vehicles which are free from defects resulting from defective material, improper manufacturing or assembly practices, and which shall meet or exceed requirements herein specified. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.

4.1.1 Contractor's inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Contractor's quality control system. The contractor shall provide and maintain an effective inspection and quality control system acceptable to the Government covering the supplies under contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor will not be restricted to the inspection station or to the method of inspection listed, provided that an equivalent control is included in the approved quality assurance procedure. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification or other applicable documents referenced herein.

4.1.3 Government verification. All quality assurance operations performed by the contractor shall be subject to Government verification at unscheduled intervals. Verification shall consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written inspection plan are being properly applied and (b) Government produce inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures or instances of poor practices which might have an effect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrective action has been taken or until conformance of product to specified criteria has been demonstrated.

4.1.4 Qualified products. When product qualification is required, the contractor inspection records shall list all components, name of suppliers, the number and date of applicable QPLs.

4.1.5 Service products. The contractor shall submit to the Government certification that products conforming to table I paragraph 3.5 of this specification will be used in servicing vehicles under contract.

4.2 First article inspection.

4.2.1 In-process inspection. The first vehicle of each model produced shall receive an in-process inspection by the Government when the vehicle is approximately 50 percent complete. This inspection shall be conducted to evaluate workmanship and materials to applicable specification. During this inspection, welding procedures will be reviewed, workmanship specimens evaluated, and the substructure examined for conformance to contractural requirements.

4.2.2 Completed vehicle inspection. The first vehicle, of each model, when completed, shall be given a preliminary inspection to determine conformance to contract and specification requirements. This inspection shall be conducted by the Government at the contractor's plant, subsequent to contractor inspection.

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4.2.3 Corrections. Corrections found necessary as a result of the above inspections, shall be made by the contractor on the vehicle.

4.2.4 Vehicle disposition. Upon completion of inspection, the vehicle shall remain at the manufacturing facility and be the last vehicle shipped on the contract. The contractor shall service and maintain the vehicle during this period in accordance with the applicable documents for care and preservation while in storage.

4.2.5 Final approval and acceptance. Final approval and acceptance by the Government of the first vehicle shall be withheld until a final determination has been made regarding conformity of the vehicle to contractual requirements including, but not limited to, workmanship and materials.

4.3 Inspection provisions.

4.3.1 Lot formation. A lot shall consist of all vehicles of one type, from an identifiable production period, submitted at one time for inspection and tests.

4.3.2 Sampling.

4.3.2.1 Examination. For the purpose of visual, dimensional and primary functional examination, a representative sample shall be selected from each lot in accordance with inspection level II of MIL-STD-105, using the following classification of defects and acceptable quality levels (AQL's) listed in tables III and IV.

Table III. Major defects AQL = 25 defects/100 units

Defect No.	Characteristics	Major defect	Method of inspection
101.	Frame	Structural, welding, and riveting defects.	Visual.
102.	Axles	Improper assembly or installation; welding defects; unusual noise.	Visual and functional.
103.	Service and parking brakes	Malfunction; unusual noise; component damage; nonconformance to holding.	Visual and functional.
104.	Automatic actuation feature	Malfunction, nonconformance	Functional.
105.	Electrical system components and circuits.	Malfunction; inoperative; lighting sequence improper damage.	Visual and functional.
106.	Suspension system components	Damage	Visual.
107.	Wheels and tires	Tire damage	Visual.
108.	Safety chains, intervehicular hose, electrical cables, tubing, wiring, and ground straps.	Damage; improper lengths	Visual and functional.
109.	Lunette eye	Casting defects; size	Visual.
110.	Trailing ability	Nonconformance	Functional.
111.	Turning ability	Nonconformance	Functional.
112.	Towing speed (highway)	Nonconformance	Functional.
113.	Operating units	Overheating	Tactile (touch)

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Table IV. Minor defects AQL = 150 defects/100 units

Defect No.	Characteristics	Minor defect	Method of inspection
201.	Wheels and tires	Improper type, size, or mounting	Visual.
202.	Electrical cables, tubing, wiring and ground straps.	Improper coded, protected, installed, or assembled.	Visual.
203.	Brake system components	Improper assembly, installation, or clearance.	Visual.
204.	Electrical system components	Improper assembly or installation	Visual.
205.	Suspension system components	Improper assembly or installation	Visual.
206.	Frame, and sheet metal	Improper installation or assembly	Visual.
207.	Lunette eye	Improper assembly or installation	Visual.
208.	Tie downs, lashing hooks, and lifting devices	Missing; improperly located; welding defects.	Visual.
209.	Reflectors	Missing; defective, improper installation.	Visual.
210.	Paint	Application and color improper	Visual.
211.	Lube fittings	Defective; missing; improperly installed.	Visual.
212.	Decals, data and instruction plates	Incomplete data; improper size or installation.	Visual.
213.	Lubrication	Improper application	Visual.
214.	Hydraulic fluid	Level and grade improper	Visual.
215.	Protective coatings	Application, coverage and materials improper.	Visual.
216.	Record forms	Missing; improperly processed	Visual.
217.	Tools	Missing; improperly processed or secured.	Visual.

4.3.2.1.1 Unclassified characteristics. All defects having no bearing on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots, or ten lots or more within a thirty day period, will be added to the minor classification of characteristics with no increase in AQL's. This deficiency may be deleted from the minor classification of characteristics when five consecutive lots are found free of the deficiency.

4.3.2.1.2 Rivet construction. Riveted construction not covered by listed specification shall conform to accepted workmanship specimens approved by the Government. Accepted workmanship specimens shall be considered the minimum acceptable standard of quality for each similar type of construction available to the term of the contract.

4.4 Tests.

4.4.1 Performance. Vehicle shall be subjected to road and performance tests specified in table V. Vehicles shall be serviced with specified products listed in table I of paragraph 3.5. Vehicle tests shall be conducted at location or locations in table V.

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Table V. Locations for tests

Title of test	Requirement paragraph	Test paragraph	Placement of manufacturer
Highway Operation	3.5.1.1	4.5.2	X
Turning ability	3.5.4	4.5.2	X
Control test	. . .	4.5.3	X
First vehicle	. . .	4.2	X

4.4.2 Five mile test. Each vehicle selected in accordance with 4.3.2 shall be operated for distance of not less than 5 miles by the contractor. Vehicles shall be completely assembled and serviced. Test course or area shall be selected by the manufacturer in the vicinity of the manufacturing facility and shall be on smooth, approximately level, hard surfaced roads. Vehicles (less payload) shall satisfactorily pass the performance requirements specified in 3.4.1.1, (highway towing speed), 3.4.3 (braking holding ability), 3.4.4 (turning ability), and 3.4.6 (wheel and tire balancing).

4.4.3 Control test. The Government shall select at random for road test by the contractor one out of each 100 vehicles submitted, but not more than 2 vehicles in any given 30-day period, for a distance of 50 miles with full or simulated payload. Vehicles selected shall be subjected to test specified in 4.4.2.

4.4.3.1 Test failure. If a vehicle fails to pass any control test specified herein, the Government shall stop acceptance of subsequent lots until objective evidence has been provided by the contractor that corrective action has been taken to eliminate the condition which caused rejection.

4.5 Preservation and packaging.

4.5.1 Lot size. A lot shall consist of level A and level C processed vehicle from an identifiable production period submitted at one time for inspection.

4.5.2 Sampling.

4.5.2.1 Examination. For the purpose of visual inspection a representative sample shall be selected from each lot in accordance with inspection level II of MIL-STD-105, using the following classification of defects listed in table VI.

Table VI. Minor defects - 4% defective

Defect No.	Area	Minor defects	Method of inspection
201.	Vehicle cleanness	Improper	Visual
202.	Disassembled parts	Improperly processed	Visual
203.	Unprotected, exposed and plated surfaces	Preservative missing, improper grade and application	Visual Visual
204.	Record forms	Missing or incomplete data, improperly processed	Visual
205.	Marking	Wrong, missing or incomplete location and size improper.	Visual

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5. PREPARATION FOR DELIVERY

5.1 Vehicle processing. Vehicle shall be processed for shipment and storage in accordance with Standard MIL-STD-281 to the extent indicated on the Vehicle Preservation Data Sheets or other implementing document, as furnished by the responsible Government activity at the time of invitation for bid (see 6.2).

6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for use by the armed services for the transportation of equipment, general cargo and supplies.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Model of vehicle to be furnished (see 1.2).

(c) Information to be shown on data plates (see 3.8).

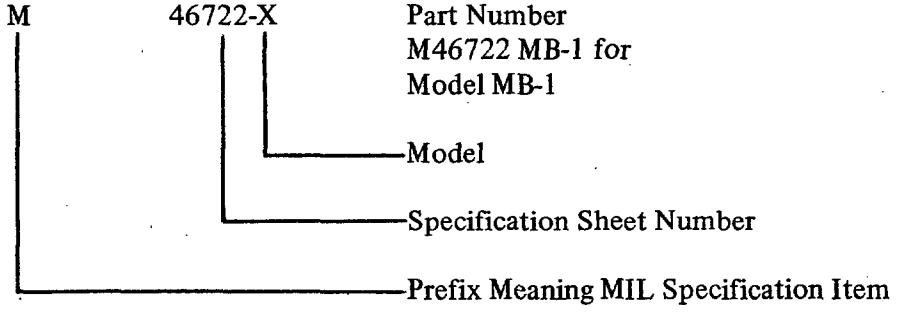
(d) Specify level of vehicle processing (see 5.1).

(e) Painting procedure and color of paint (see 3.6).

6.3 Recycled Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.4 Supersession data. This specification supersedes MIL-G-8299A (USAF); and that portion of MIL-T-45320A (Ord) pertaining to chassis, trailer, 4-wheel XM11322, XM456 and MIL-C-46722.

6.5 Part numbering system. Military Specification part numbering system is as follows:



6.6 Prime movers.

Chassis		Preferred prime mover
MB-1	M44	2 1/2 ton truck series
XM113E2	M44	2 1/2 ton truck series
XM456	M40	5 ton truck series

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6.7 Design responsibility. Performance requirements are established to provide criteria for determining conformance of the vehicle to specified requirements, including workmanship and materials. The contractor shall not assume design responsibility unless specifically directed by the rocuring agency through engineering order.

6.8 International Standardization Agreement. Certain provisions of this specification are the subject of international standardization agreement NATO STANAG 2805-A (Edition 3). When amendment, revision, or cancellation of this specification is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

6.9 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval for inspection and approval of the first article.

6.10 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Air Force - 99

Preparing Activity:

Air Force - 84
(Project 2330 - 0933)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-46722A	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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